Airfoil Selection for Two RC Aircraft

Aircraft A: maximum payload with stable flight. Aircraft B: maximum maneuverability (turns, rolls, loops).

Numbers below are approximate 2D values at Re ≈ 150,000-250,000, typical for small RC wings. Low-Re data varies a lot with surface finish, turbulence and source. Verify the finalists in XFOIL, XFLR5 or airfoiltools.com at your own Reynolds number before committing.

1. Shortlists

Aircraft A (payload, stable)

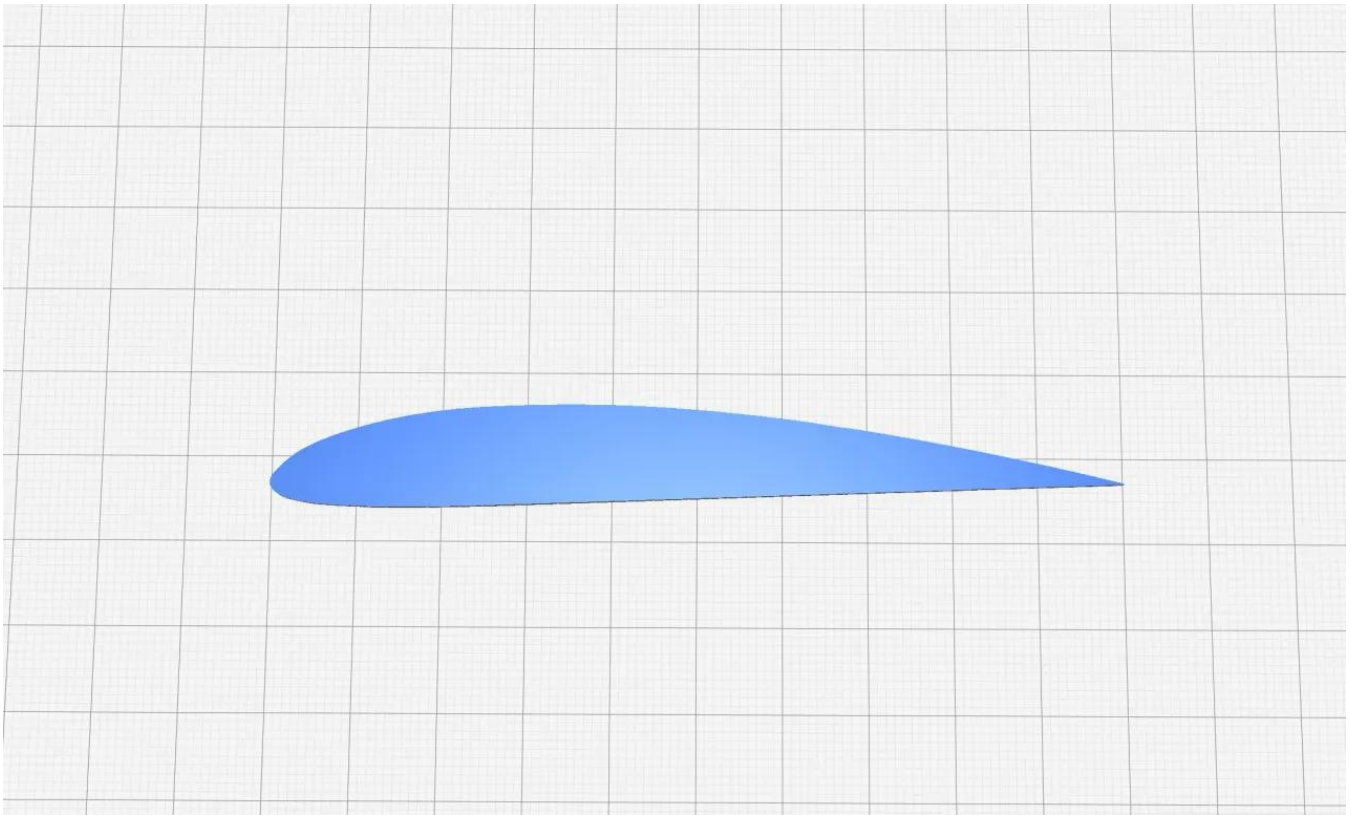
Airfoil	Camber	Thickness	Cl max	Cd min	Max L/D	Stall
Clark Y	~3.5% (flat bottom)	11.7%	~1.3	~0.013	~35-45	Very gentle
NACA 4412	4%	12%f	~1.4	~0.012	~40-50	Gentle
Eppler E423	~9.5%	~12.5%	~2.0	~0.016	~50-70	Moderate, fairly docile
Selig S1223	~8.7%	~12%	~2.2	~0.02	~50-60	Sharper, high drag off-design

Aircraft B (maneuverability)

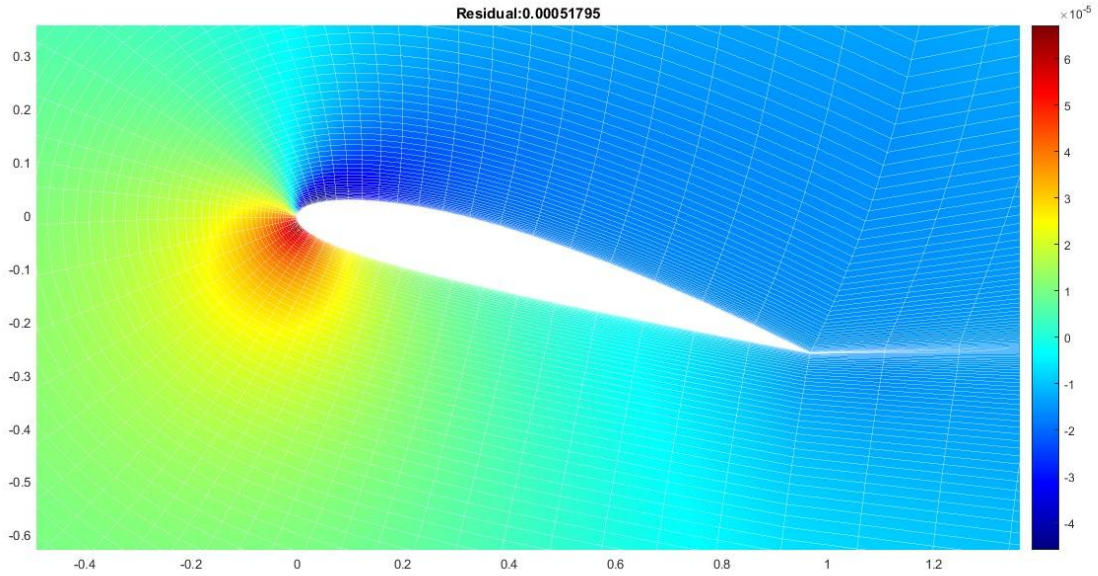
Airfoil	Camber	Thickness	Cl max	Cd min	Max L/D	Stall
NACA 0009	0%	9%	~0.9	~0.011	~25-30	Abrupt (thin-airfoil type)
NACA 0012	0%	12%	~1.0-1.1	~0.012	~25-30	Progressive
NACA 0015	0%	15%	~1.1	~0.014	~20-25	Very gentle

Airfoil profiles

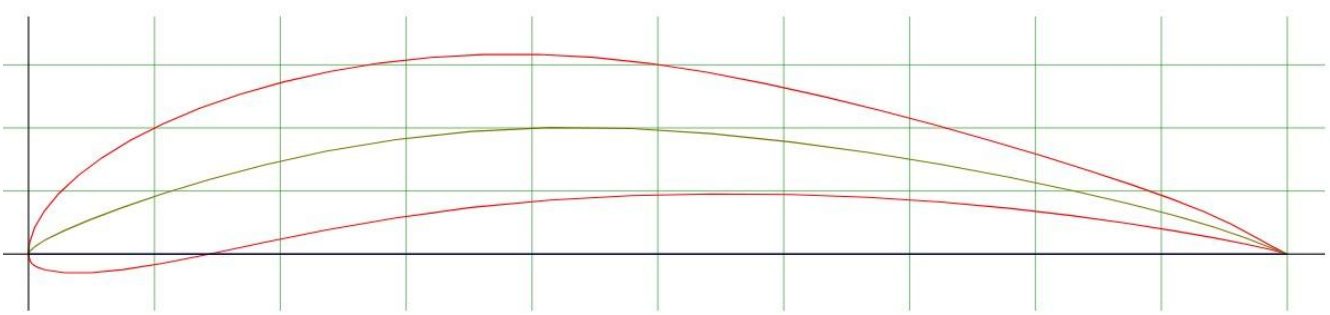
1. **Clark Y**



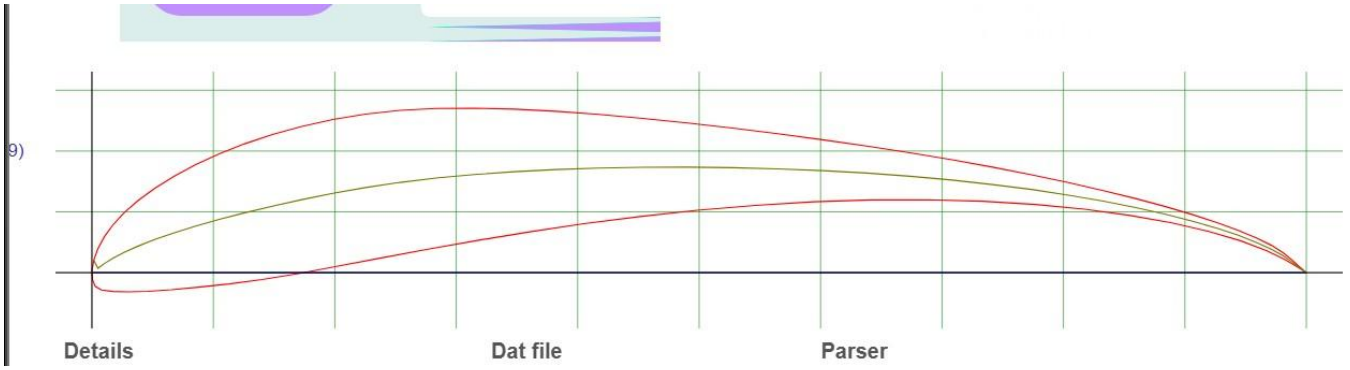
2. NACA 4412



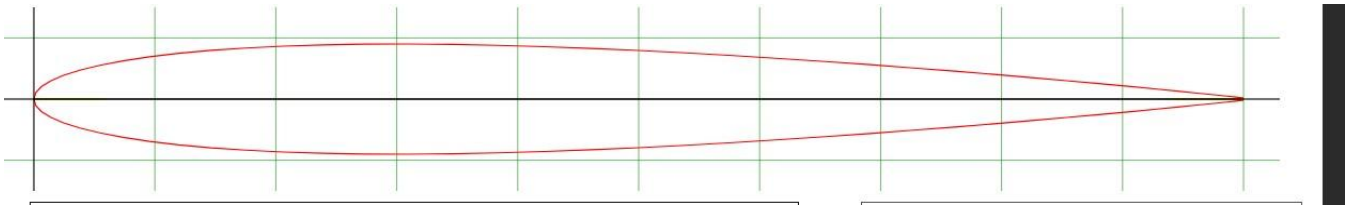
3. Eppler E423



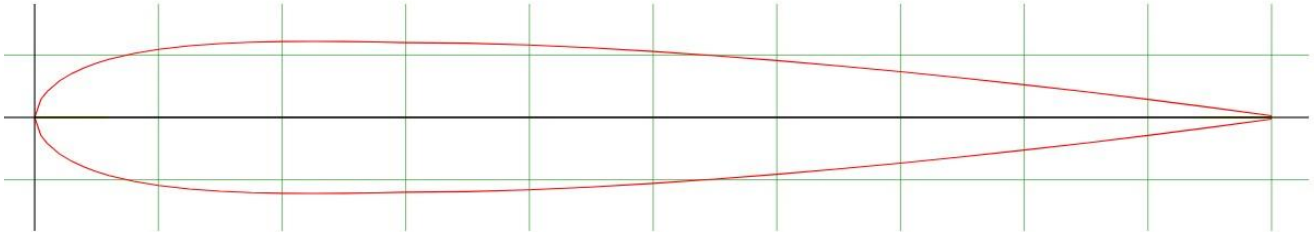
4. **Selig S1223**



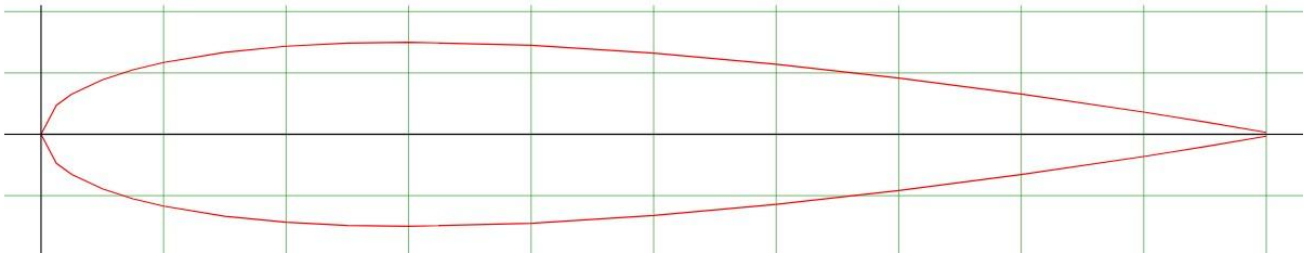
5. **NACA 0009**



6. **NACA 0012**



7. **NACA 0015**



2. Selections

Aircraft A: Eppler E423

Payload capacity is set largely by how little wing area and speed you need to carry the weight, and that depends on Cl max (stall speed ∝ 1/√Cl max). E423's Cl max of about 2.0 gives roughly 25-30% lower stall speed than Clark Y at the same wing loading, or the same stall speed with about 50% more weight per unit area. Its high L/D at high Cl also means efficient climb and cruise while loaded. It is 12%+ thick, so the spar can be deep and light for carrying heavy loads.

The costs are a strongly negative pitching moment (Cm around -0.2), so you need a generous tail, forward CG and some download on the stabilizer. It also has more drag at low Cl and a less forgiving

stall than Clark Y. If the pilot is inexperienced or the wing must be flat-bottomed for easy building, Clark Y or NACA 4412 is the safe fallback, with a larger wing to compensate.

Aircraft B: NACA 0012

Symmetric sections fly identically upright and inverted, which is essential for loops, rolls and knife-edge. Pitching moment is near zero and the center of pressure stays near 25% chord, so trim barely changes as Cl changes. The stall is progressive, which helps in snap maneuvers, and 12% thickness is strong enough for a light structure. NACA 0009 has slightly less drag but a more abrupt stall and a weaker spar. NACA 0015 is very forgiving but slower and draggier.

3. How Geometry Drives Performance

- **Camber** shifts the lift curve upward, giving more Cl at zero angle of attack and a higher Cl max. It also increases nose-down pitching moment and drag at low Cl, and makes inverted flight inefficient.
- **Thickness** raises Cl max up to about 12-15%, because the leading-edge radius delays separation. Beyond that it adds drag. It also gives structural depth (spar stiffness goes roughly as depth cubed) and internal volume.
- **Leading-edge radius** controls stall type. A blunt nose gives gradual trailing-edge separation, while a sharp one gives abrupt leading-edge stall.
- **Position of maximum camber and thickness** moves the center of pressure and pitching moment. Aft camber gives more lift but a larger nose-down moment.
- **Reynolds number** matters more here than on full-size aircraft. Below about Re 200k, laminar separation bubbles can raise drag and soften published performance. Surface finish and a turbulator or trip strip can help.

4. Can One Airfoil Do Both?

Not optimally. The needs conflict directly:

Need	A (payload)	B (aerobatic)
Camber	High, for Cl max	Zero, for symmetry
Pitching moment	Large, tolerated by big tail	Near zero, for trim consistency
Inverted flight	Poor	Equal to upright
Cruise Cl	High (~1.0-1.5)	Low (~0.2-0.5)

A high-camber section like E423 flies inverted badly and needs constant elevator input in maneuvers. A symmetric NACA 0012 needs a much larger wing or higher speed to carry the same payload. Compromises exist, such as Clark Y, NACA 4412, or semi-symmetrical sections like NACA 2412, or 2-3% camber foamy-style sections. These carry moderate loads and do mild aerobatics. They suit a trainer or

sport plane, but they are second-best at both extremes. If one aircraft must do both, choose a semi-symmetrical 10-12% section and accept the compromise.

5. Airfoil Versus Whole-Aircraft Factors

The airfoil sets the 2D limits: Cl max, drag, stall type, pitching moment and structural depth. Other factors often matter more:

- **Wing loading (W/S)** dominates stall speed, turn rate and payload. Aircraft A wants low W/S, such as 6-10 oz/ft² for a loaded trainer-class design, via large area. Aircraft B wants low-to-moderate W/S for tight turns but enough for penetration.
- **Aspect ratio and planform** drive induced drag and roll inertia. A has high AR (7-9) for efficiency; B has lower AR (5-6) and often a rectangular or lightly tapered wing for crisp roll response and a stall that starts at the root. The higher the aspect ratio the greater the lift .

$$\textbf{Aspect Ratio} = \textbf{Span}^2 \ / \ \textbf{Wing area}$$

- **Control surfaces** determine roll rate and pitch authority. B needs large, full-span ailerons (often with differential), a big elevator and rudder, and low control-system slop. Airfoil choice cannot compensate for undersized controls.
- **Tail volume and CG** give A its stability: big stabilizer, forward CG and dihedral. B uses smaller tail moments, an aft CG and little or no dihedral. Forward CG increase longitudinal stability and makes stall recovery easier it creates a heavier nose down pitch that the tail must fight with a larger downward force ,increasing drag and stall speed.

- **Thrust-to-weight** matters hugely for B: ≥1:1 for vertical maneuvers and rapid recovery. A needs enough thrust for takeoff with payload and climb, typically 0.5-0.7:1 with a high-torque, large-diameter prop.

- **Structure** includes spar sizing, wing joiner and landing-gear strength, which set the usable payload and G-limits. Thickness helps but does not replace good design. •**Wing twist (washout)** and tip stall behavior are aircraft-level choices that modify the airfoil's stall character.

In short, pick the airfoil to fit the mission's Cl range and stall behavior, then design wing area, controls, tail and power around it. Wing loading and control authority usually matter more than which similar-class airfoil you choose.